

CASE STUDY

To help boost sales, Adio, the data analyst at Adventure Works, is tasked with investigating sales, marketing, and customer data. Adio is instructed to create and share a data report with visualizations, based on the insights, patterns, and trends he uncovers during the analysis. Adio knows that creating visualizations with stakeholder experience in mind contributes to improved stakeholder understanding and decision-making. To understand the needs, preferences, and expectations of the stakeholders that form his audience (stakeholder experience), Adio engages in the following process:

Step 1: Identifying stakeholders

The first step in the stakeholder experience process is to determine which stakeholders have an interest in the data analysis and visualizations. Adio identifies the executives, marketing team, and product managers as stakeholder groups for his current project.

Step 2: Defining stakeholder goals

Aware that these stakeholder groups will have diverse needs, preferences, and expectations, Adio moves on to the next step—understanding the different stakeholders' goals, priorities, and requirements. Through consultation with each group of stakeholders, Adio determines their respective interests in the sales data analysis:

- The group of executives is focused on improving the overall performance of the company and is interested in high-level insights that can drive strategic decisions related to their competitors, products, and customer marketing.
- The marketing team is interested in insights into marketing campaign effectiveness and how to improve conversion rates from their website and social media sites to increase sales.
- The product managers want to understand customer behavior, such as product popularity, preferences of different customer segments or groups, and the profitability of each of these segments.

Step 3: Choosing the right visualization type

Adio can now choose visualization types that are tailored to the stakeholders' goals and needs. Bar charts, line charts, and pie charts are common visualization types, but there are many others to consider. The visualizations you choose will depend on the type of data you're working with. You also have to think about which visualization types will best communicate the insights your stakeholders need while being visually appealing and easy to understand. For example:

- For executives, a dashboard with high-level metrics and key performance indicators (or KPIs) such as revenue and profit margins may be the right choice.
- The marketing professionals may find a conversion funnel that tracks the progress of customers from the stage of visiting marketing channels to the final stage of product purchases more useful. A bar chart to compare the effectiveness of different marketing channels may also be an appropriate choice.
- For the product managers, a map that visualizes the distribution of customer segments may be suitable.

Step 4: Designing with stakeholder experience in mind

Next, you need to design the visualizations with stakeholder experience in mind, ensuring visualizations are meaningful to each group. This means designing visualizations that answer stakeholder questions, are visually appealing, and easy to read, navigate and understand. When designing his visualizations, Adio keeps in mind that:

- The executives prefer visualizations that are concise, easy to understand, and convey key takeaways quickly.
- The marketing professionals are interested in visualizations that can help them identify trends, patterns, and opportunities for improvement.
- The product managers value visualizations that can help them identify gaps, opportunities for growth, and potential issues with specific products.

Step 5: Making visualizations interactive

You can enhance stakeholder engagement, exploration, and understanding of data insights by adding interactive features to your visualizations. For example, you can add filters and sorting options, or explanations that appear when stakeholders hover over different parts of a visualization. Adio uses Power BI to create interactive data visualizations that the stakeholders can explore and interact with. For example, he adds filtering and sorting options. That way, when Adio shares the report, the marketing team can view data by marketing channels or campaigns. Similarly, the product team can view sales distribution data by customer segment or product category.

Step 6: Testing and iterating

Before sharing the visualizations with stakeholders, Adio tests them with a small group of users that represent his target audience. Focus groups involve selecting and recruiting a diverse and representative sample of individuals from the different stakeholder groups and then conducting the focus group sessions. In this final step, you should gather and analyze feedback and make any necessary adjustments to ensure that your visualizations effectively communicate the intended insights. In doing so, you ensure that the stakeholders' needs are met.

Answer questions about stakeholder experience

1. What is the primary goal of data visualization in the data analysis process?
2. Briefly define stakeholder experience in the context of data analysis and visualization.
3. Explain two reasons why stakeholder experience is important when creating data visualizations.
4. List the six steps in the stakeholder experience process.
5. How can you identify the goals and preferences of different stakeholder groups when designing visualizations?
6. What is one challenge you think Adio may face in designing data visualizations that meet the needs of the different stakeholders?

7. Briefly discuss what you need to consider when choosing the right visualization type for stakeholders.
8. What is the purpose of making visualizations interactive?
9. A focus group finds a visualization Adio designed difficult to understand. Adio adds more interactive features and updates the design. Which step in the process is Adio engaging in?
10. In no more than two sentences, discuss how you think stakeholder experience can contribute to improving business outcomes through data-driven decision-making.

SOLUTION

1. What is the primary goal of data visualization in the data analysis process?

The primary goal of data visualization is to present data insights in a clear, concise, and visually engaging manner to help stakeholders quickly understand key information and make informed decisions.

2. Briefly define stakeholder experience in the context of data analysis and visualization.

Stakeholder experience refers to understanding and addressing the needs, goals, and preferences of stakeholders when creating and sharing data visualizations to ensure clarity, relevance, and impact in communication.

3. Explain two reasons why stakeholder experience is important when creating data visualizations.

- It ensures that the visualizations are relevant and meaningful, allowing stakeholders.
- It increases stakeholder engagement and satisfaction, leading to better communication and collaboration in the decision-making process.

4. List the six steps in the stakeholder experience process.

1. Identifying stakeholders
2. Defining stakeholder goals
3. Choosing the right visualization type
4. Designing with stakeholder experience in mind
5. Making visualizations interactive
6. Testing and iterating

5. How can you identify the goals and preferences of different stakeholder groups when designing visualizations?

You can identify goals and preferences by consulting directly with stakeholders through interviews, meetings, or surveys, and asking targeted questions about their roles, decision-making needs, and the types of insights they value most.

6. What is one challenge you think Adio may face in designing data visualizations that meet the needs of the different stakeholder

Adio may struggle to balance the diverse requirements and data literacy levels of different stakeholder groups while keeping the visualizations clear and not overly complex.

7. Briefly discuss what you need to consider when choosing the right visualization type for stakeholders.

You need to consider the type of data being analyzed, the specific insights each stakeholder group needs, and their ability to interpret different visual formats. The goal is to choose a format that is both intuitive and effective in communicating key messages.

8. What is the purpose of making visualizations interactive?

Interactive visualizations allow stakeholders to explore data in more detail, filter

information based on their needs, and better understand specific trends, leading to deeper insights and more informed decisions.

9. A focus group finds a visualization Adio designed difficult to understand. Adio adds more interactive features and updates the design. Which step in the process is Adio engaging in?

Adio is engaging in **Step 6: Testing and iterating**.

10. In no more than two sentences, discuss how you think stakeholder experience can contribute to improving business outcomes through data-driven decision-making.

By tailoring data visualizations to stakeholder needs, organizations can ensure clearer understanding and faster action on key insights. This leads to more effective strategies, higher stakeholder engagement, and ultimately better business outcomes.